Kernix 0.11.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

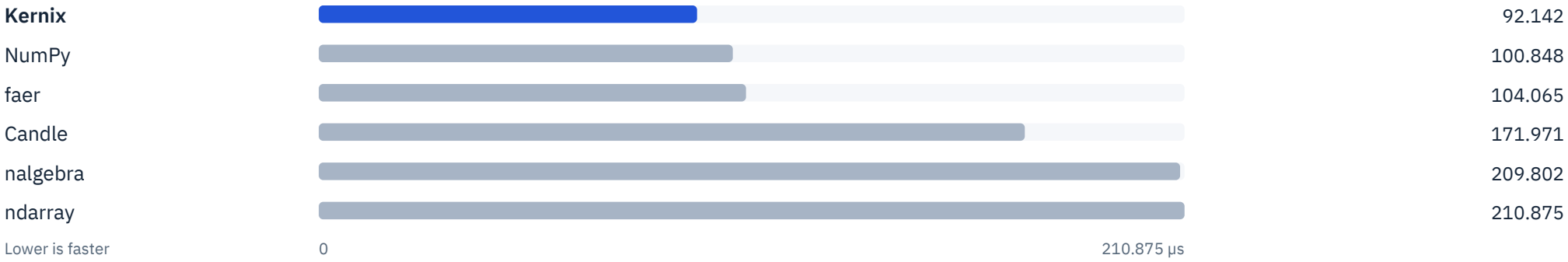
35 workloads

119 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



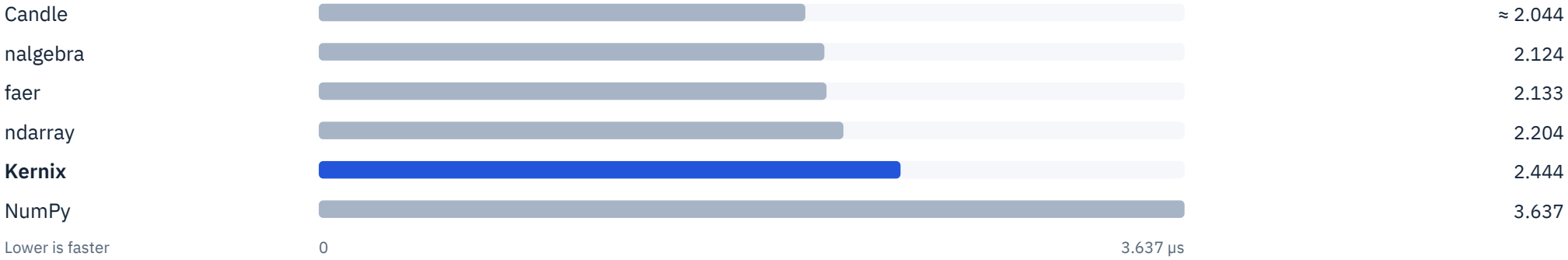
Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

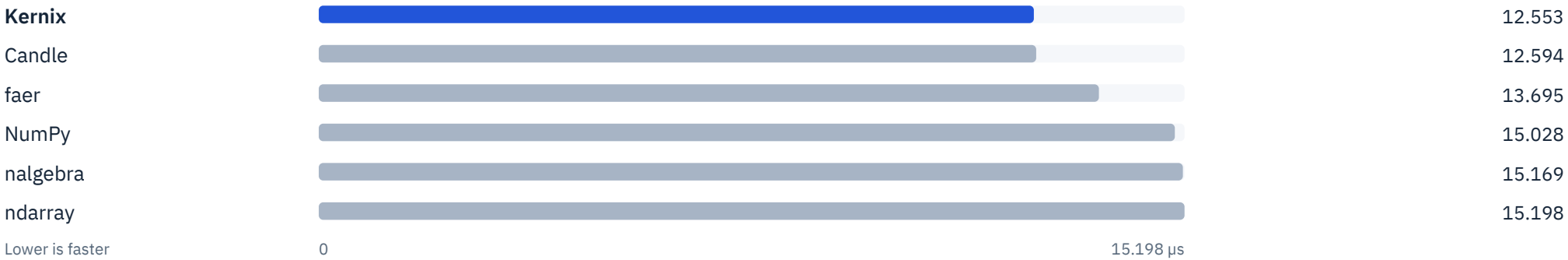
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



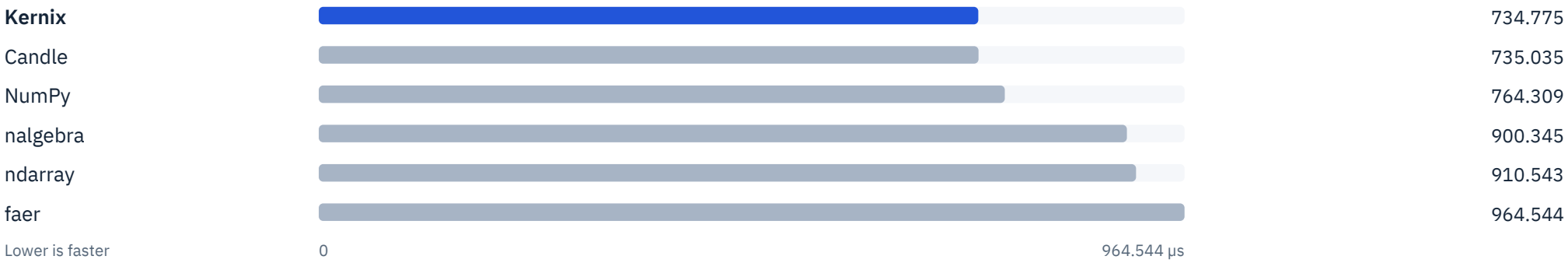
Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

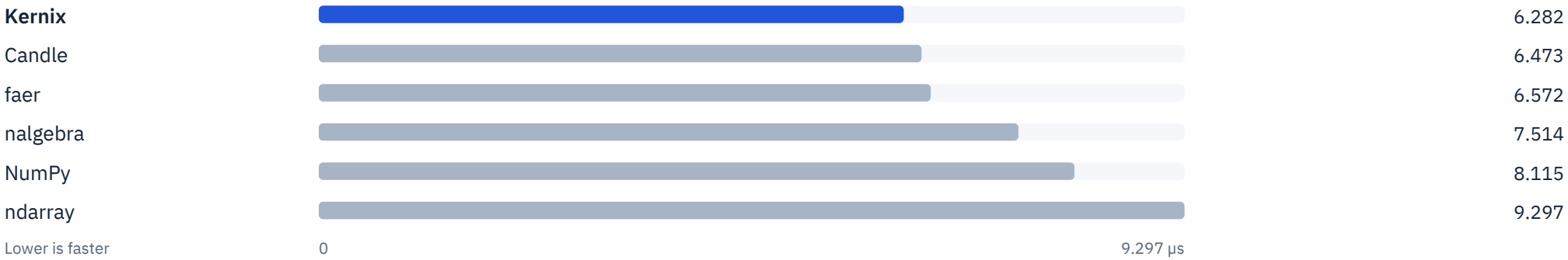
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



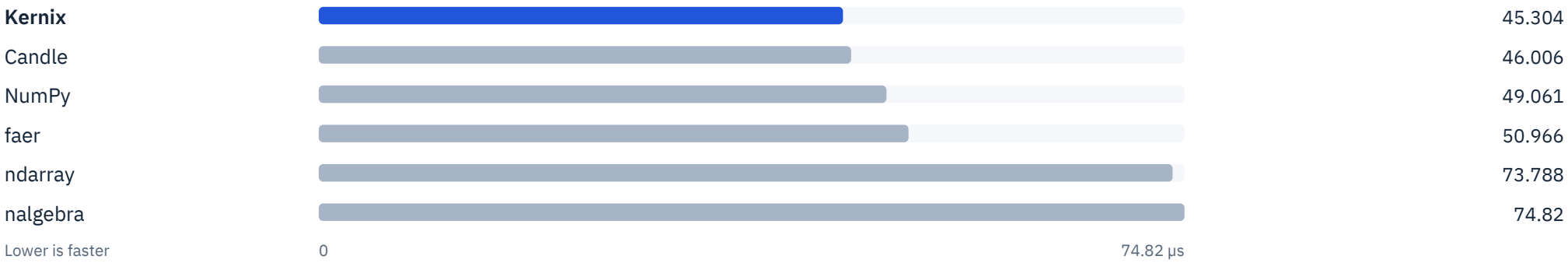
Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

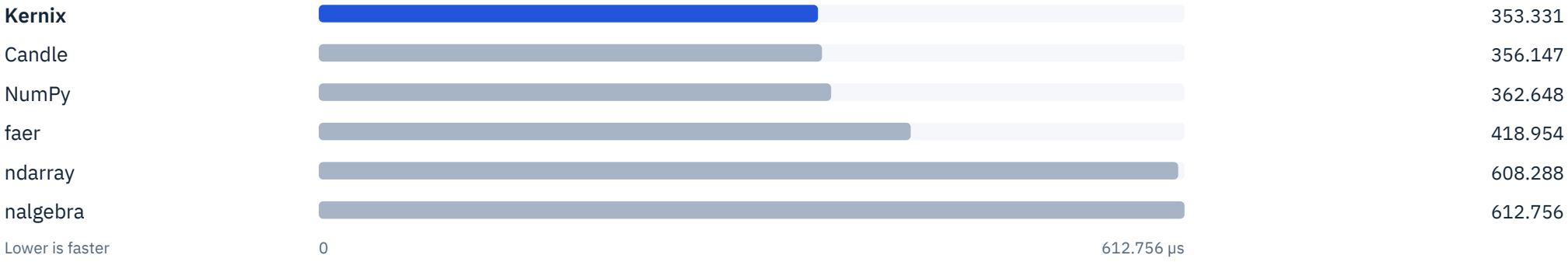
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

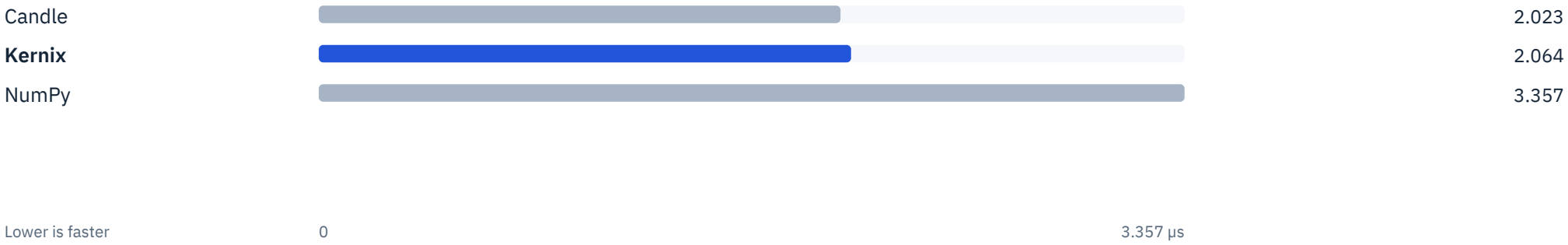
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

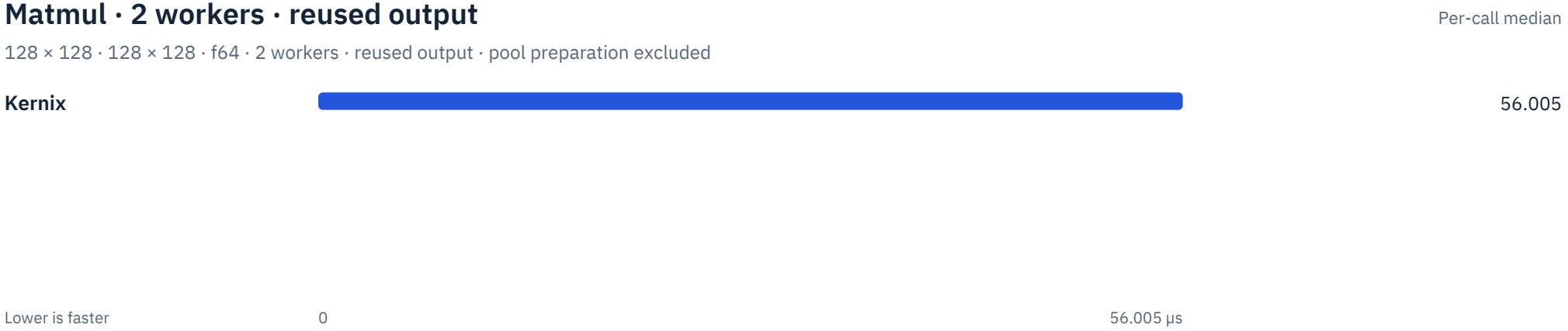
Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



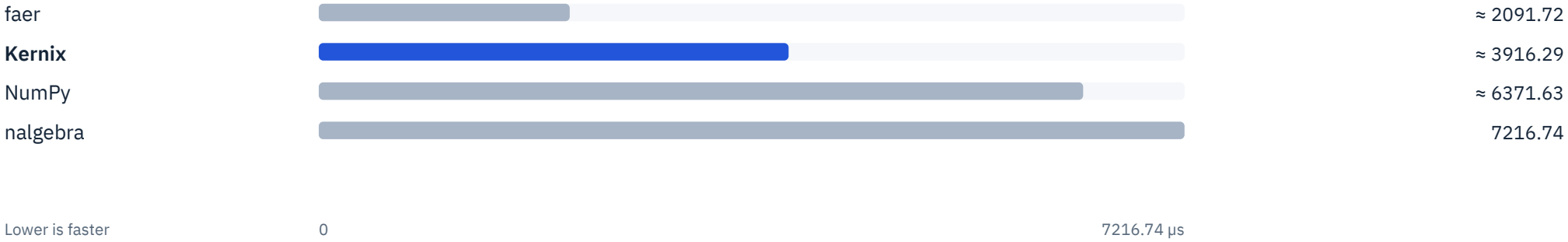
Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

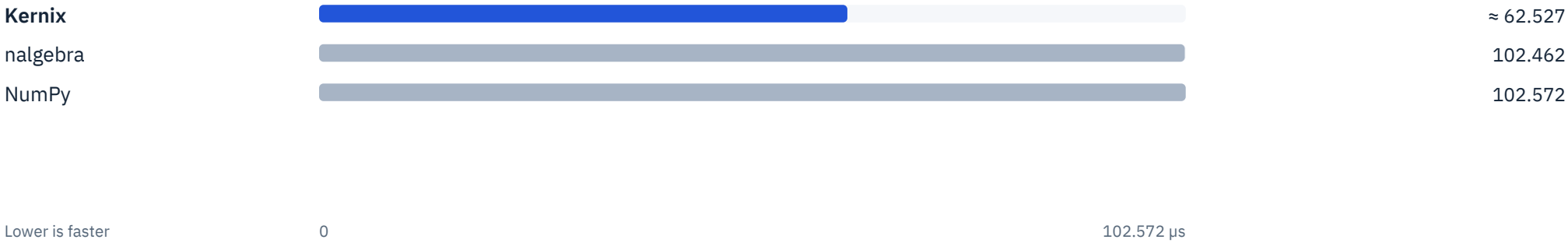
Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



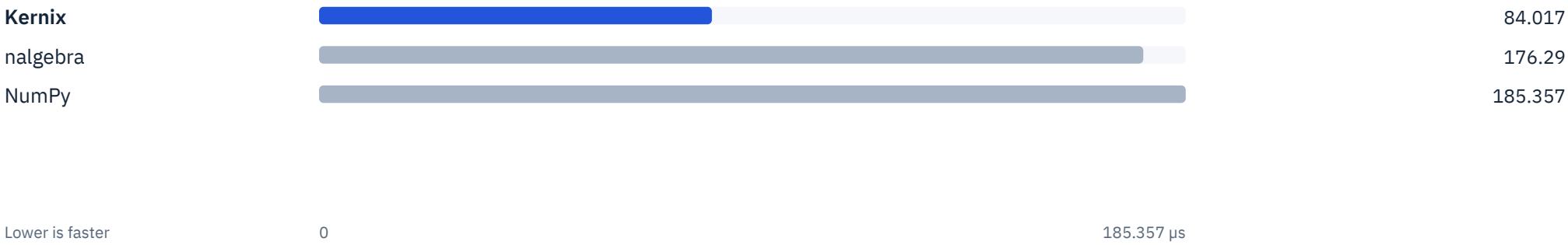
Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

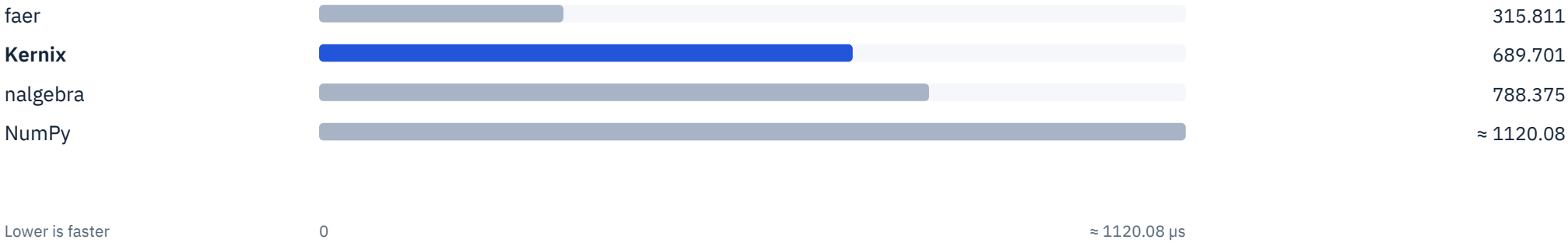
128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

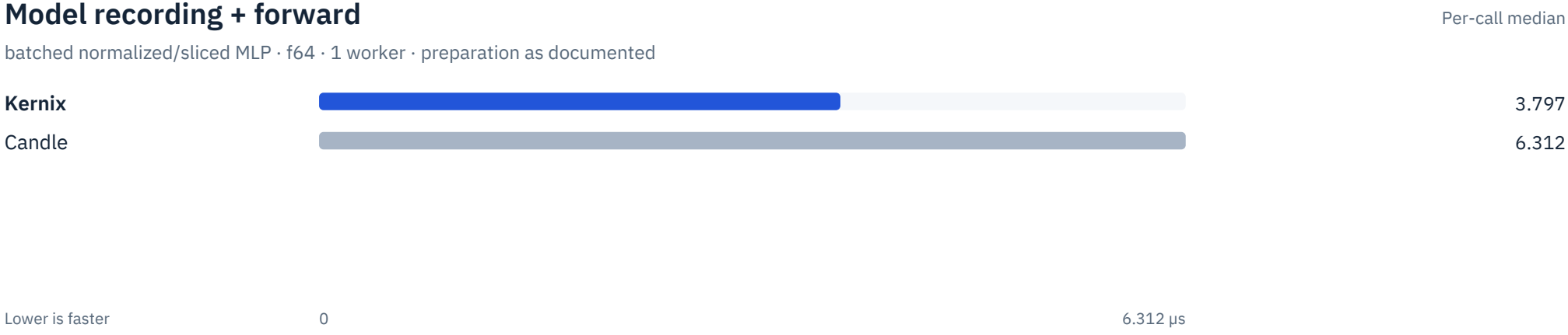
Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



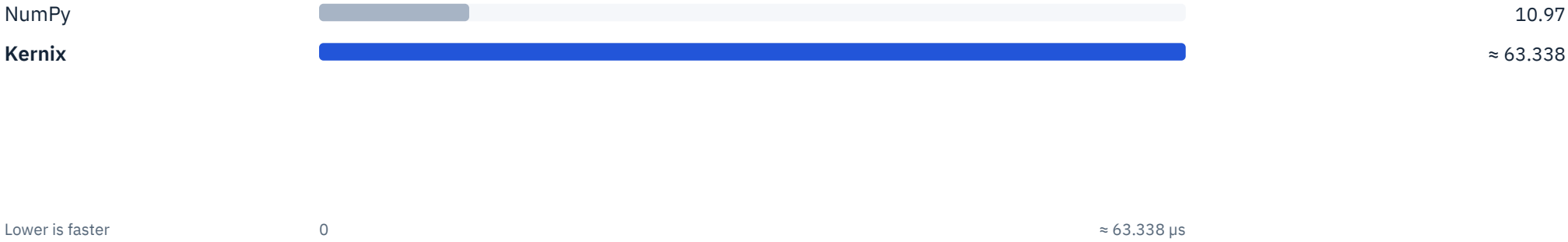
Kernix 0.11.0 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Per-call median

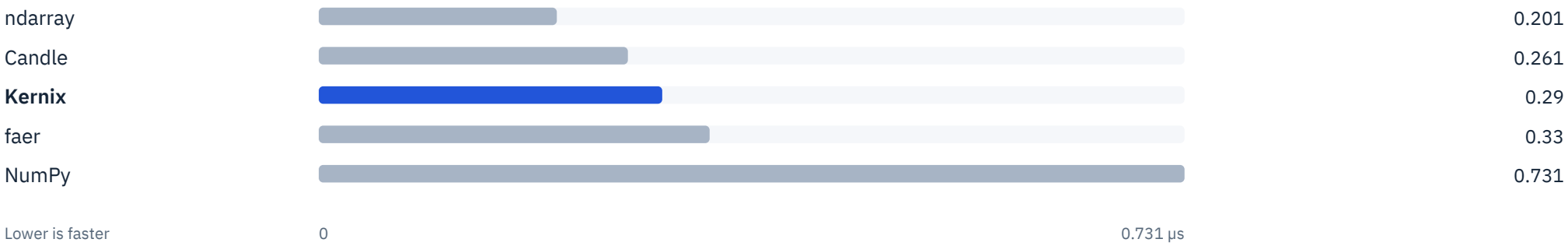


Kernix 0.11.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	101600	101.6
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	93455	93.455
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	49212	49.212
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	45486	45.486
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2043.9999999999998	2.0439999999999998
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2204	2.204
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2124	2.124
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2133	2.133
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	3637	3.637
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2444	2.444

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	15198	15.198
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	15028	15.028
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	12594	12.594
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	13695	13.695
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	15169	15.169
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	12553	12.553
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	100848	100.848
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	210875	210.875
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	92142	92.142
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	104065	104.065

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	209802	209.802
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	171971	171.971
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	764309	764.309
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	735035	735.035
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	900345	900.345
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	910543	910.543
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	964544	964.544
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	734775	734.775
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	9297	9.297
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	8115	8.115

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	6572	6.572
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6282	6.282
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	6473	6.473
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7514	7.514
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	74820	74.82
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	50966	50.966
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	46006	46.006
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	73788	73.788
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	45304	45.304
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	49061	49.061

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	612756	612.756
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	362648	362.648
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	608288	608.288
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	418954	418.954
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	353331	353.331
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	356147	356.147
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3035	3.035
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	4148	4.148
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3046	3.046
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2023	2.023

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	3357	3.357
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	2064	2.064
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	16130	16.13
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	56005	56.005
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	391522	391.522
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	31189.000000000004	31.189000000000004
Cholesky factorization / nalgebra 512 × 512 · f64	7216740	7216.74
Cholesky factorization / NumPy 512 × 512 · f64	6371631	6371.631
Cholesky factorization / faer 512 × 512 · f64	2091724	2091.724
Cholesky factorization / Kernix 512 × 512 · f64	3916288.0000000005	3916.2880000000005

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 96 × 32 · f64	42901	42.901
Column-pivoted QR / nalgebra 96 × 32 · f64	77114	77.114
Column-pivoted QR / Kernix 96 × 32 · f64	78647	78.647
Column-pivoted QR / nalgebra 256 × 64 · f64	765482	765.482
Column-pivoted QR / faer 256 × 64 · f64	387886	387.886
Column-pivoted QR / Kernix 256 × 64 · f64	799817	799.817
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	102462	102.462
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	102572	102.572
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	62527.00000000001	62.52700000000001
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	185357	185.357

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	84017	84.017
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	176290	176.29
Thin QR / faer 96 × 32 · f64	55404	55.404
Thin QR / NumPy 96 × 32 · f64	69831	69.831
Thin QR / Kernix 96 × 32 · f64	67856	67.856
Thin QR / nalgebra 96 × 32 · f64	59661	59.661
Thin QR / faer 256 × 64 · f64	426127	426.127
Thin QR / nalgebra 256 × 64 · f64	571269	571.269
Thin QR / Kernix 256 × 64 · f64	756375	756.375
Thin QR / NumPy 256 × 64 · f64	543668	543.668

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	25237	25.237
Cholesky factorization / faer 64 × 64 · f64	14307	14.307
Cholesky factorization / NumPy 64 × 64 · f64	18294	18.294
Cholesky factorization / nalgebra 64 × 64 · f64	21009	21.009
Cholesky factorization / NumPy 128 × 128 · f64	112010	112.01
Cholesky factorization / nalgebra 128 × 128 · f64	124702.99999999999	124.70299999999999
Cholesky factorization / faer 128 × 128 · f64	58660	58.66
Cholesky factorization / Kernix 128 × 128 · f64	169076	169.076
Cholesky factorization / NumPy 256 × 256 · f64	1120085	1120.085
Cholesky factorization / faer 256 × 256 · f64	315811	315.811

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 256 × 256 · f64	788375	788.375
Cholesky factorization / Kernix 256 × 256 · f64	689701	689.701
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4629	4.629
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	10299	10.299
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	5431	5.431
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3676	3.676
Model inference / Candle batched normalized/sliced MLP · f64	4328	4.328
Model inference / Kernix batched normalized/sliced MLP · f64	2936	2.936
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3797	3.797
Model recording + forward / Candle batched normalized/sliced MLP · f64	6312	6.312

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Kernix prepared graph → fresh gradients · f64	4729	4.729
Model backward / Candle prepared graph → fresh gradients · f64	17042	17.042
Parameter update / Kernix transactional SGD, fixed gradients · f64	792	0.792
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	63337.99999999999	63.33799999999999
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	10970	10.97
Row softmax / Candle 256 × 64 · f64	84929	84.929
Row softmax / ndarray 256 × 64 · f64	83135	83.135
Row softmax / Kernix 256 × 64 · f64	93435	93.435
Row softmax / NumPy 256 × 64 · f64	120315	120.315
Create zeros / faer 64 × 64 · f32	330	0.33

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Candle 64 × 64 · f32	261	0.261
Create zeros / NumPy 64 × 64 · f32	731	0.731
Create zeros / Kernix 64 × 64 · f32	290	0.29
Create zeros / ndarray 64 × 64 · f32	201	0.201
Create ones / Candle 64 × 64 · f32	451	0.451
Create ones / ndarray 64 × 64 · f32	351	0.351
Create ones / NumPy 64 × 64 · f32	2084	2.084
Create ones / Kernix 64 × 64 · f32	430	0.43
Create ones / faer 64 × 64 · f32	391	0.391